

Highlights

Remote RGB Cardiac-Rate Information Under Spectral Masking

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- Cardiac-rate-consistent spectral information often remained present in remote RGB when the blind dominant peak selected a slower component.
- In recovery recordings, CHROM retained a near-reference candidate in most windows, while blind selection was correct in only 3/20 and 4/20 windows.
- Measurement-side masking was reproducible in declared simulations, but the physical nuisance mechanism remains unidentified.

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Abstract

Remote photoplethysmography is often evaluated by whether the largest spectral peak matches reference heart rate. In this single-participant, single-camera multimodal study, we examined the distinction between cardiac-rate-consistent spectral information being present and being selected as the dominant component. We used 8th-generation iPad RGB recordings with Polar H10 ECG, contact MAX30102 photoplethysmography, and exploratory chest audio. Across stationary and post-exercise recordings in this acquisition setting, remote RGB frequently contained a component near the independently measured cardiac rate, including while that rate changed substantially. Blind dominant-peak selection nevertheless often selected slower competing structure. In 150 overlapping windows from a stationary 15-take corpus, CHROM contained a candidate within 5 bpm of H10 in 122 windows, while its blind winner was within 5 bpm in 88. In two post-exercise recovery recordings, CHROM retained a candidate near the independently measured cardiac rate in 16/20 and 17/20 windows, while the blind winner was correct in only 3/20 and 4/20. A temporal continuity selector and a fixed spatial heterogeneity selector did not materially resolve candidate identity. A six-interval multimodal run provided a clear example at 124.5 bpm: CHROM contained a 123 bpm candidate while Y and POS selected approximately 47 bpm. Mechanistic observation-model screens showed that measurement-side masking is expressible without altering the prescribed latent cardiovascular trajectory, but did not identify the physical nuisance mechanism. These findings support an observation and identifiability interpretation rather than a validated blind heart-rate estimator.

Keywords: remote photoplethysmography, rPPG, cardiac rate, observability, spectral masking

1. Introduction

Remote photoplethysmography (rPPG) converts small camera-observed colour changes into a temporal signal related to cardiovascular pulsation. Contact PPG provides the physiological background for this measurement problem [1]; contactless multi-wavelength imaging and ambient-light camera plethysmography established the early lineage [2, 3]. The usual evaluation compresses each recording to one estimated heart rate and compares it with an ECG reference. This is useful for deployment-oriented benchmarking, and recent reviews place classical projections, learned spatiotemporal models, datasets, and translation challenges within the same broader field [4, 5]. However, it conflates two questions:

1. Is there a component near the independently measured cardiac rate in the camera signal?
2. Does a camera-only estimator identify that component as the dominant or selected component?

Here, “observability” is used operationally to mean that information near the independently measured cardiac rate is present in an observed signal; it does not invoke the formal rank-based observability definition used in control theory. The distinction matters when the observation contains several structured temporal processes. A cardiac-rate-consistent component may survive while a slower illumination, geometry, processing, or other observation-side component has greater spectral power. Conversely, agreement between a camera peak and ECG does not by itself establish skin specificity, because narrow-band camera structure can occur in non-skin controls.

This study therefore treats rPPG as an observation and identifiability problem. We separate frozen camera outputs from H10-guided diagnostic analyses, use contact and acoustic channels as independent context, and test whether simple camera-only temporal or spatial selectors can recover a component that is present but subdominant.

2. Methods

2.1. Measurement architecture

The primary remote signal was a cheek-region iPad RGB trace. Camera-side channels included luminance-like Y, POS, and CHROM [6, 7]; exploratory comparators included GREEN, PBV, OMIT, PCA, and ICA. PBV, blind source separation, and learned spatiotemporal methods represent established

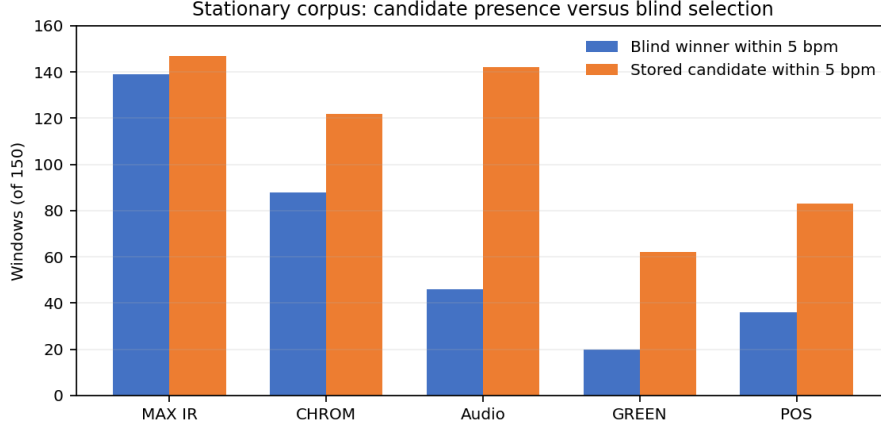


Figure 1: Stationary corpus: candidate presence exceeded blind dominant selection.

alternative priors [8, 9, 10, 11, 12]. A Polar H10 supplied ECG and heart-rate reference data. A MAX30102 supplied contact red and infrared PPG. A BOYA BY-M1 lavalier microphone replaced one stethoscope earpiece and was coupled to that earpiece tubing, supplying exploratory chest audio. In later hardware tests an AD8232 ECG front end and MAX30102 were sampled by the same ESP32 clock.

The iPad interval was the master interval for multimodal comparisons. H10 and ESP32 streams were cropped by host wall time. Camera outputs were generated before H10 was revealed for estimator evaluation. H10-guided candidate analyses were performed only after camera spectra were fixed and are labelled diagnostic.

2.2. Datasets

The principal physical corpus comprised 15 stationary iPad/H10 intervals, of which 14 had complete reference coverage and one was retained with partial coverage. The same recordings supplied 150 overlapping diagnostic windows. Two separate 90-second post-exercise recovery recordings followed 30 jumping jacks; each was analysed using 12-second windows advanced every 4 seconds. A later six-interval multimodal recording comprised three resting iPad intervals, 30 jumping jacks, and three recovery intervals, with continuous H10, ESP32 ECG/PPG, and audio acquisition.

2.3. Frozen estimators and diagnostics

Each RGB trace was timestamp-interpolated to a uniform 30 Hz grid, divided by its recording mean, and linearly detrended. Welch PSD used `nperseg=min(512,N)`, `noverlap=min(256,N//2)`, and linear detrending. The frozen stationary search band was 0.7–3.0 Hz (42–180 bpm), matching the predeclared Welch protocol. The recovery selector used its separately frozen 30–180 bpm (0.5–3.0 Hz) candidate range. A candidate was a stored Welch spectral bin in the applicable band, ordered by power; no H10 information entered candidate generation. The stationary diagnostic list retained the five strongest stored bins, while the recovery selector retained the ten strongest. These are bins rather than separated local maxima, so adjacent bins may represent one broad lobe. The local-background score used the median PSD in the candidate’s ± 0.25 Hz neighbourhood, excluding the candidate and immediate neighbouring bins; edge neighbourhoods were clipped to the search band. Equal scores within $1e-9$ used the lowest frequency/component index. Diagnostic candidate presence was assessed only after revealing H10, with ± 5 bpm tolerance for the stationary summary and ± 10 bpm for recovery windows. The dynamic analyses retained this distinction:

- *Reference-blind dominant peak*: the strongest candidate selected without using H10 in each interval or window. Camera-only refers specifically to the selectors that operate on camera data alone.
- *Cardiac-rate-consistent candidate*: a stored candidate close to the independently measured H10 rate, evaluated post hoc.

Two development-only camera selectors were tested on the recovery recordings. The temporal selector connected candidate frequencies across adjacent windows with a fixed transition penalty and restart rule. The spatial selector used fixed 4x4 tiles and a predeclared combination of whole-ROI power and tile-amplitude heterogeneity. Spatial redundancy and ROI choice are established rPPG concerns [13, 14, 15]; this selector was intentionally a limited diagnostic rather than an exhaustive spatial method. Neither selector used H10, contact PPG, or audio during selection.

2.4. Observation-model simulation

The synthetic branch used a fixed cardiovascular trajectory and fixed optical transfer. Synthetic camera-physiology data and tissue optical modelling provide relevant methodological precedents [16, 17, 18, 19]. Declared observation-side perturbations were applied without fitting amplitudes or

Table 1: Stationary 150-window summary. Candidate presence is H10-guided diagnostic analysis.

Signal	Reference-blind dominant peak within 5 bpm	Candidate within 5 bpm
MAX30102 IR	139	147
CHROM	88	122
Audio	46	142
GREEN	20	62
POS	36	83

seeds to physical recordings. Screens included common illumination, channel gain drift, spatial mixing, motion-colour mixing, exposure integration, explicit half-frequency competition, and stochastic ROI, illumination, ISP, and motion-coupled mechanisms. The simulation question was capability: can cardiac-rate information survive while blind ranking fails? A final fixed-nuisance trajectory screen used a 90-to-125-to-94 bpm trajectory and a single unchanged 45 bpm nuisance component across the full run.

3. Results

3.1. Stationary component presence and dominant selection

In the 150-window stationary analysis, MAX30102 infrared PPG was within 5 bpm of H10 in 139 windows and contained a cardiac-region candidate in 147. CHROM was within 5 bpm as a blind winner in 88 windows, but contained a candidate within 5 bpm in 122. Audio contained a cardiac-region candidate in 142 windows but selected it blindly in only 46. The difference between candidate presence and blind selection was therefore substantial, particularly for CHROM and audio.

The complete-reference stationary take analysis gave CHROM dominant-peak agreement in 10/14 takes and CHROM cardiac-region candidate presence in 12/14. GREEN and POS were substantially less reliable as blind dominant estimators in this corpus.

3.2. Dynamic recovery recordings

The two 90-second recovery recordings produced a substantial H10 trajectory. In the first, H10 began near 120 bpm and settled near 95–100 bpm; in

Dynamic recovery: reference, contact PPG, and blind CHROM

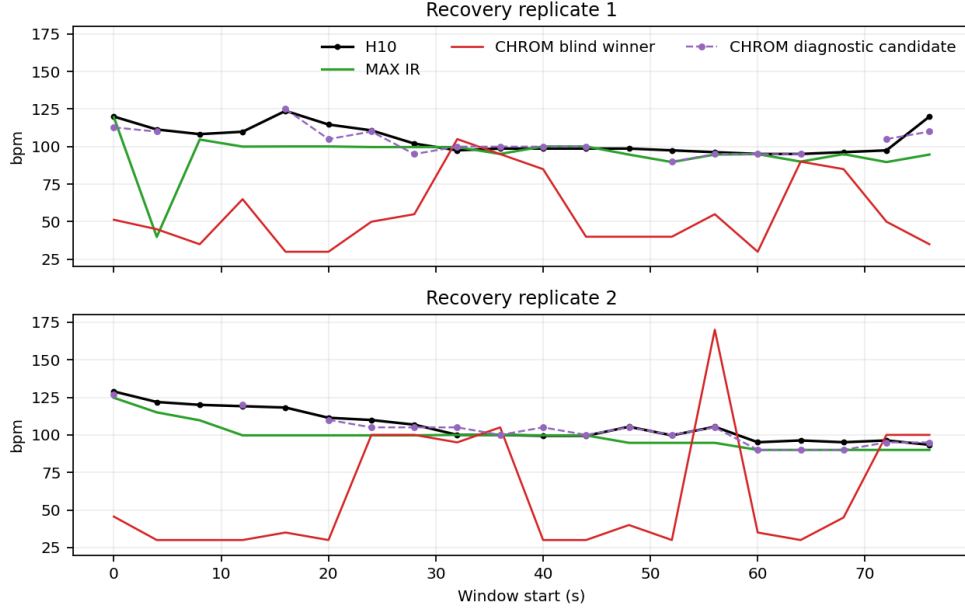


Figure 2: Two recovery recordings. H10, contact infrared PPG, blind CHROM winner, and the H10-guided diagnostic CHROM candidate are shown separately. The dashed diagnostic trace is not a blind estimator output.

the second it began near 129 bpm and fell through approximately 120, 110, and 100 bpm. Contact infrared PPG broadly followed these changes.

CHROM retained a candidate near the changing H10 rate in 16/20 windows in recovery replicate 1 and 17/20 in replicate 2. The blind CHROM winner was within 10 bpm in only 3/20 and 4/20 windows, respectively. Early windows provide the clearest example: H10 and contact PPG were near 115–125 bpm, CHROM contained a candidate in that region, and a slower component near 30–50 bpm won the blind ranking.

3.3. Six-interval multimodal run

The first recovery interval of the six-interval run had H10 at 124.5 bpm, AD8232 at 109.0 bpm, and MAX30102 red at 118.7 bpm. CHROM contained a 123 bpm component, while Y and POS selected 47.1 and 48.8 bpm. Later recovery intervals produced H10 rates of 103.1 and 94.2 bpm; CHROM followed at 99.6 and 93.8 bpm, while POS again selected a slow component in the final

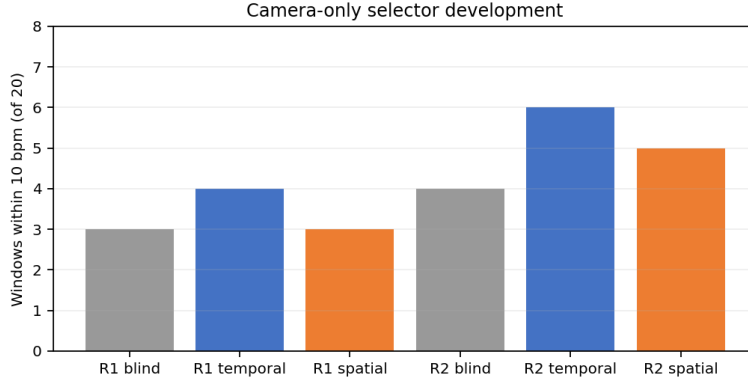


Figure 3: Camera-only selector development. Neither tested selector materially rescued identification.

interval.

This interval-level result is important because the heart rate changed between recordings. The remote component near the high-rate reference was not merely a stationary coincidence at approximately 90 bpm, although its physiological specificity is not established by frequency agreement alone.

3.4. Camera-only selector development

The temporal continuity selector improved blind agreement from 3/20 to 4/20 in recovery replicate 1 and from 4/20 to 6/20 in replicate 2. The spatial heterogeneity selector produced 3/20 versus 3/20 and 5/20 versus 4/20, respectively. The selectors usually followed the same slow component as the ordinary whole-ROI winner. These development results were frozen without further tuning.

3.5. Hardware controls

The hardware channels were used as independent context rather than as a second estimator-development programme. Across a five-take host-time-mapped stationary batch, AD8232 rate estimates differed from contemporaneous H10 by 0.2–0.7 bpm. Same-ESP32-clock ECG-to-infrared ordering was observed in 73 of 83 coarse pairings. MAX30102 red and infrared and chest audio were conditionally recoverable. Audio repeatedly contained cardiac-rate structure and approximately doubled event-rate structure; beat-level PCG timing and S1/S2 identification were not attempted because the audio clock

Table 2: Interpretation of the observation-model capability screen. These are qualitative capabilities, not comparative evidence that one nuisance is more plausible.

Synthetic condition	Candidate survives	Blind ranking can fail
Fixed baseline	yes	yes
Channel gain drift	yes	condition-dependent
Spatial mixing	yes	yes
Motion-colour mixing	yes	yes
Exposure integration	yes	condition-dependent
Explicit low-frequency competitor	yes	positive-control demonstration

was not hardware-locked to the ECG. Detailed engineering results are retained in the Supplementary Materials.

3.6. Simulation screens

The dynamic observation-model screen reproduced the qualitative pattern of cardiac candidate survival with blind ranking failure under several declared perturbations. An explicit low-frequency competitor was a positive control for spectral masking. The stochastic screen found that tested ROI and ISP mechanisms preserved the cardiac peak, while noncommon illumination and combined mechanisms could overwhelm all channels without reproducing the selective Y/POS hierarchy. Thus the measurement layer is capable of expressing masking, but these screens do not identify the real nuisance mechanism. In the final fixed-nuisance trajectory screen, the cardiac candidate remained present in all 20 windows; Y remained nuisance-dominant while POS and CHROM remained cardiac-dominant. The complete cardiac-dominant to masked to recovered sequence was therefore not automatic under that fixed geometry, although projection divergence was expressed.

4. Discussion

The central result is physical:

Across stationary and post-exercise recordings, remote RGB often contains a spectral component near the independently measured cardiac rate, including while that rate changes substantially. Blind dominant-peak selection nevertheless frequently chooses a different, slower component.

This formulation is deliberately narrower than calling every H10-adjacent component a physiological pulse. Empty-scene controls showed that narrow-band camera structure can occur without skin, and phase-locking statistics were not specific enough to separate cheek from empty-scene controls in short recordings. Frequency agreement therefore establishes cardiac-rate consistency, not skin-specific origin.

Recent work has also examined elevated heart rates, difficult illumination, ROI dependence, and demographic coverage, making those factors relevant comparison points rather than explanations of the present recordings [20, 15, 21, 22]. The recovery experiments strengthen the operational observability result because the cardiac-rate-consistent component moved across a broad range while the blind camera winner often remained elsewhere. The result is not that the iPad always estimates heart rate; it is that the camera can preserve information that a simple dominant-peak rule cannot reliably identify.

The selector negatives are also informative. They are negative algorithm results, not evidence that no better selector could ever exist. Temporal smoothness is not enough because nuisance components can be smooth. The tested spatial heterogeneity statistic is not enough because spatial structure alone does not identify the source. More elaborate selection should be evaluated only prospectively; tuning these rules on the current recordings would convert a mechanistic result into retrospective optimisation.

The contact and hardware channels clarify but do not erase the observation problem. They are supporting controls rather than the primary result. AD8232 and H10 provide a strong rate reference, while contact PPG and audio are themselves conditional measurement channels. Their failures reinforce the broader point that cardiovascular information can be present without dominating a particular projection or channel.

5. Limitations

The recordings are from one subject and one camera/acquisition configuration. Overlapping windows are not independent observations. H10-guided candidate presence is diagnostic rather than blind estimator performance. Empty-scene H10-frequency structure limits physiological specificity claims. AD8232 clipping limits morphology and beat-timing interpretation. Audio and H10 were not hardware-clocked, so PCG timing and S1/S2 identification remain unresolved. The simulation screens are simplified capability experiments, not fits to measured camera parameters. No conclusion about

pigmentation, universal rPPG identifiability, PAT/PTT, or blood pressure is made.

6. Conclusion

Remote RGB frequently contains cardiac-rate-consistent information, including during substantial heart-rate change, but that information is not reliably identifiable by spectral dominance or the tested temporal and spatial heuristics. Mechanistic simulation shows that such masking is compatible with the observation layer, while the physical origin of the competing component remains unresolved. The next work is consolidation, figure preparation, and prospective testing of any future selector on new data; the present acquisition corpus should remain frozen.

Ethics and data availability

The measurements analysed in this evaluation were collected from the author alone; no other participants' data were used. The underlying raw and derived measurements are not publicly available and are not provided with this article.

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